



Monitoring Report

CARBON OFFSET UNIT (CoU) PROJECT



Title: 50 MW Solar Project by Sunshakti Solar Power Projects Pvt Ltd in Telangana

Version 1.3

Date 17/02/2026

First CoU Issuance Period: 07 years, 10 months, 18 days

Monitoring Period: 02/11/2017 to 31/12/2024



Monitoring Report (MR)
CARBON OFFSET UNIT (CoU) PROJECT

Monitoring Report	
Title of the project activity	50 MW Solar Project by Sunshakti Solar Power Projects Pvt Ltd in Telangana
UCR Project Registration Number	542
Version	1.3
Completion date of the MR	17/02/2026
Monitoring period number and duration of this monitoring period	Monitoring Period Number: First Issuance Duration of this monitoring Period: 02/11/2017 to 31/12/2024
Project participants	M/S Sunshakti Solar Power projects private Limited
Host Party	India
Applied methodologies and standardized baselines	ACM0002-Consolidated baseline methodology for grid-connected electricity generation from renewable sources -Version 22.0
Sectoral scopes	01 Energy industries (Renewable/Non-Renewable Sources)
Actual amount of GHG emission reductions for this monitoring period	2017: 5584 CoUs (tCO ₂ eq)
	2018: 86896 CoUs (tCO ₂ eq)
	2019: 88105 CoUs (tCO ₂ eq)
	2020: 70243 CoUs (tCO ₂ eq)
	2021: 65924 CoUs (tCO ₂ eq)
	2022: 83660 CoUs (tCO ₂ eq)
	2023: 82475 CoUs (tCO ₂ eq)
	2024 : 71737 CoUs (tCO ₂ eq)
Total:	554,625 CoUs (tCO ₂ eq)

SECTION A. Description of project activity

A.1. Purpose and general description of project activity >>

a) Purpose of the project activity and the measures taken for GHG emission reductions >>

The project activity titled '50 MW Solar Power Plants in Telangana by M/s Sunshakti Solar Power projects private Limited' is located in Pedayyipally, in the South side of Hyderabad in Telangana with latitude and longitude are 18.22° N and 78.32° E respectively. The project site Pedayyipally is located in the sun belt of ~ 5.5-5.6 kWh/m² per day.

Name	Plant capacity	Site	Commissioning Date
M/S Sunshakti Solar Power projects private Limited	50MW	Pedayyipally	02/11/2017

The power produced by the 50MW is evacuated at 132 KV level with Interconnection Point at 220/132 KV Kamareddy SS. The generated Power is being sold to Telangana State Northern Power Distribution Company Limited (TSNPDCL).

Name of the project	Interconnection Substation	Injection Voltage	Interconnection Point	Type of Project	Capacity of the Project (MW)
M/s. SunShakthi Solar Power Projects Pvt. Ltd.	220/132 KV SS Kamareddy	132 KV	132 kV side at the 220/132 KV SS Kamareddy	Solar Photovoltaic	50 MW

Purpose of the project activity:

The project activity displaces an equivalent amount of electricity which would have otherwise been generated by fossil fuel dominant electricity grid. The estimated lifetime of the project activity is considered as 25 years for solar technology. In the Pre- project scenario the entire electricity, consumed by the customers or delivered to the grid by, would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources. The generation of power from solar photovoltaics is a clean technology as there is no fossil fuel-fired or no GHG gases are emitted during the process. A photovoltaic module consists of several

photovoltaic cells connected by circuits and sealed in an environmentally protective laminate, which forms the fundamental building blocks of the complete PV generating unit. Several PV panels mounted on a frame are termed PV Array. Thus, project activity leads to a reduction the GHG emissions as it displaces power from fossil fuel-based electricity generation in the regional grid. The technological details have been provided in Section A.4. Since the project activity generates electricity through solar energy, a clean renewable energy source it will not cause any negative impact on the environment and thereby contributes to climate change mitigation efforts.

The Crediting Period of the Project activity is 7 years from 02/11/2017 to 31/12/2024 in which total Actual electricity generation is **631,307 MWh** and total GHG emission reduction is **554,625tCO₂e**. with the average plant load factor of **21.64%**. This annual average net electricity generation and annual average GHG emission reductions are with application of degradation factor of 0.5% from second year.



Satellite View of Sun Shakti Solar Project

b) Brief description of the installed technology and equipment>>

Solar PV modules: – Solar PV modules convert solar radiation directly into electricity through the photovoltaic effect in a silent and clean process that requires no moving parts. The photovoltaic
© Universal CO₂ Emission And Offset Registry Private Ltd

effect is a semiconductor effect whereby solar radiation falling onto the semiconductor PV cells generates electron movement. The output from a solar PV cell is direct current (DC) electricity. A PV power plant contains many cells connected together in modules and many modules connected together in strings to produce the required DC power output.

Inverters – Invertors are required to convert the DC electricity to alternating current (AC) for connection to the utility grid. Many modules in series strings and parallel strings are connected to the inverters

Step-up transformers: – The output from the inverters generally requires a further step-up in voltage to reach the AC grid voltage level. The step-up transformer takes the output from the inverters to the required grid voltage (33 kV)

Module mounting systems: Fixed mounting systems keep the rows of modules at a fixed tilt angle while facing a fixed angle of orientation for maximising the energy incident on the collector plane. The optimum tilt angle is generally between 10° and 35°, facing true south.

c) Relevant dates for the project activity (e.g. construction, commissioning, continued operation periods, etc.)>>

Provide the duration of the crediting period corresponding to the monitoring period covered in this monitoring report.

UCR Project ID: UCR 542

Project Start Date: 02/11/2017

Commissioning Date: 02/11/2017

d) Total GHG emission reductions achieved or net anthropogenic GHG removals by sinks achieved in this monitoring period>>

The total GHG emission reductions achieved in this monitoring period is as follows:

Summary of the Project Activity and ERs Generated for the Monitoring Period	
Start date of this Monitoring Period	02/11/2017
Carbon credits claimed up to	31/12/2024
Total ERs generated (tCO _{2eq})	554,625 tCO _{2eq}
Leakage	0

e) Baseline Scenario>>

As per the approved consolidated methodology ACM0002 Version 22, if the project activity is the installation of a new grid-connected renewable power plant/unit, the baseline scenario is the following:

“If the project activity is the installation of a Greenfield power plant, the baseline scenario is electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources to the grid”.

A.2. Location of project activity>>

Country: India

The site is located in Pedayyipally, Kamareddy District in the South side of Hyderabad in Telengana with latitude and longitude are 18.22° N and 78.32° E respectively. The site is situated at the height of about 490 m above MSL (Mean Sea Level).

Details of Latitude & Longitude of Individual machines have been Given below: -

Name	Plant capacity	Site	Geo-coordinate
M/S Sunshakti Solar Power projects private Limited	50MW	Pedayyipally	18.22° N and 78.32° E

The location of the project site has been shown below:



3. Parties and project participants >>

Party (Host)	Participants
India	M/S Sunshakti Solar Power projects private Limited

A.4. References to methodologies and standardized baselines >>

SECTORAL SCOPE - 01 Energy industries (Renewable/Non-renewable sources)

TYPE I- Renewable Energy Projects

CATEGORY- ACM0002-Consolidated baseline methodology for grid-connected electricity generation from renewable sources -Version 22.0

A.5. Crediting period of project activity >>

Type- Renewable Energy Projects

Start date- 02/11/2017

Length of the crediting period corresponding to this monitoring period: 07 years, 10 months, 18 days

A.6. Contact information of responsible persons/entities >>

Contact Person- Lokesh Jain

Email- lokesh.jain@viviidgreen.com

Phone no- 91 89208 56146

Address- Sri Krishna Complex, New Link Road, Opp. Laxmi Industrial Estate, Andheri (West), Mumbai - 400053

SECTION B. Implementation of project activity

B.1. Description of implemented registered project activity >>

a) Provide information on the implementation status of the project activity during this monitoring period in accordance with UCR PCN>>

The project consists of ground mounted photo voltaic solar plant with aggregated installed capacity of 50 MW. The plant was commissioned on several dates by the respective authority of government of Telangana. The project generates clean energy by utilizing the solar Radiations.

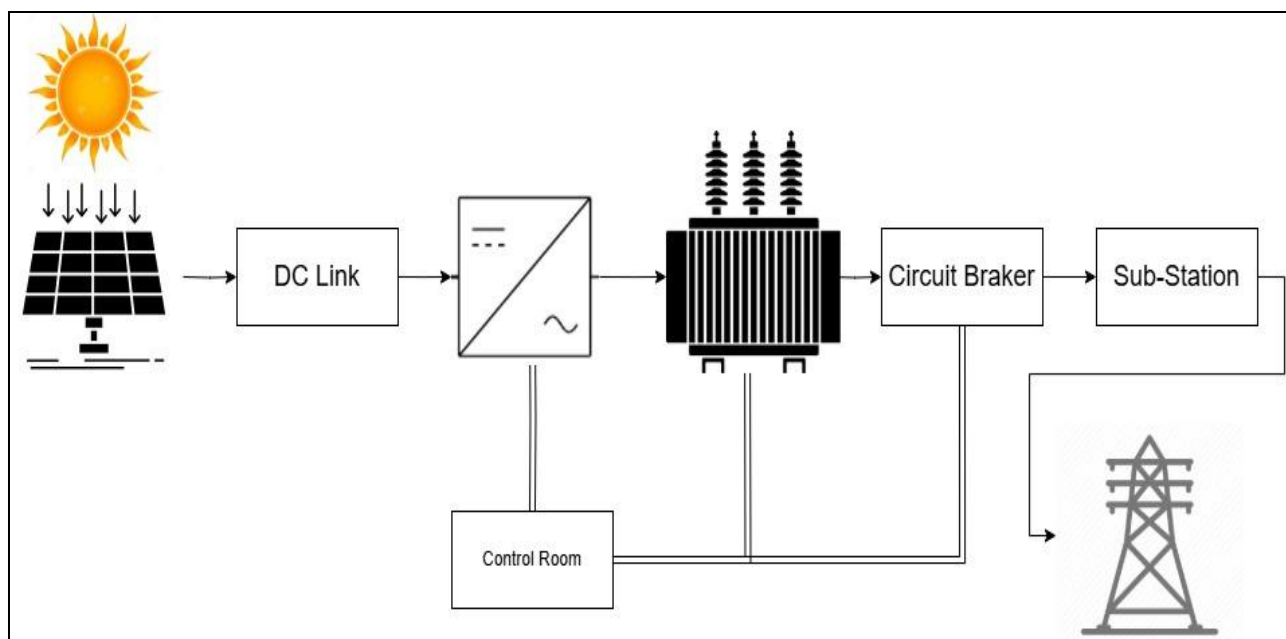
The applied technology is considered to be one of the most environment friendly technologies available as the operation of the Solar photovoltaic does not emit any GHGs or any other harmful gases unlike the operation of conventional power plants. Photovoltaic module consists of several photovoltaic cells connected by circuits and sealed in an environmentally protective laminate, which forms the fundamental building blocks of the complete PV generating unit. Several PV panels mounted on a frame are termed as PV Array.

The project activity has used the reliable and proven technology to ensure that an environmentally safe and sound technology has been implemented.

b) For the description of the installed technology(ies), technical process and equipment, include diagrams, where appropriate>>

PV modules generate direct current (DC) electricity, and it is necessary to convert into alternating current (AC) and adjust the voltage levels before powering equipment designed for normal main supply and before interfacing with grid. Conversion would be achieved by using an electronic inverter and the associated control and protection devices. All these components of the systems are termed as power conditioning unit. After conversion from DC to AC, it would be exported to the grid. In case the plant needs DC electricity, the energy generated from SPV array would be stored in batteries and can be used as and when required.

This 50 MW grid connected solar PV power plant at Pedayyipally would comprise of the following main components.



PV Array

The solar PV system relies on PV modules as its foundational components, with each module converting sunlight into DC electricity through its integrated solar cells. These crystalline silicon (Si) cells are connected in series and hermetically sealed within the module, ensuring durability and optimal performance in various weather conditions. Encased in an anodized aluminum frame, the cells are protected by a high-transmission toughened glass superstrate and a tedlar/polyester rear cover, with EVA (UV stabilized polymer) sheets encapsulating them. Electrical output is managed through a PVC module terminal box at the module's rear. Our plant exclusively uses crystalline modules, chosen for their efficiency and availability, with each module generating 20-35 V and 215-335 Wp DC power depending on the specific rating and solar irradiance. In the plant's design, modules are connected in series to form strings, precisely matching the inverter's input voltage, and multiple strings are then connected in parallel to create the solar array, fulfilling the plant's overall power requirements.

A typical example of total number of modules (Tier 1) required for the proposed capacity and its specifications (with all relevant standards) are as under:

Sl. No.	Head	Value
1.	Type of solar cell	Poly-crystalline
2.	Peak power rating	320 W _p
3.	PV modules in series/string	20
4.	Voltage at maximum power	37.56 V
5.	Current at maximum power	8.52 A
6.	Dimensions	1956 mm x 991 mm x 45 mm
7.	Weight	26.0 kg
8.	Frame	Anodized aluminum

Mounting Structure

The design of the mounting structure (single axis tracker) would be such that the tracker axis would be placed at N-S direction and the inclination of tracker is 0°, and rotation angle +58° and –58° (with backtracking). Each structure would be capable of supporting cluster of modules.

Power Conditioning Unit

Solar PV modules generate DC power, which is converted into 50 Hz 3-phase AC power at grid voltage for grid connectivity. PV modules would be arranged in suitable number of clusters and each cluster will have suitable number of inverters of efficiency of >98.6% in maximum power point (MPP) voltage range to convert generated DC power of each module to 50 Hz 3-phase AC power. Inverters will operate on maximum power point tracking (MPPT) mode to ensure maximum output from the PV modules/array. The inverters would have power measurement sensors, which would turn on the inverters when output power from solar modules exceeds threshold value. The system automatically starts in the morning and begins to export power to the grid, provided there is sufficient solar radiation and in the evening when output power from the modules falls below the threshold level then inverter would turn off. In the event of grid disturbances, the inverter would be disconnected from the grid and reconnected automatically when the grid stabilizes. When the exported power is lower than a threshold for a pre-determined time the inverter turn to sleep mode and disconnect from the grid. Total harmonic distortion of the inverters would be less than 3%.

Parameters

Sl. No.	Head	Value
1.	Continuous rating	43 kW
2.	MPPT operating voltage range	200 V ~ 1,000 V
3.	Output frequency	50 Hz ± 5%
4.	Maximum DC voltage	1100
5.	IP protection	565

b) Power Quality

To monitor and control the parameters of DC/AC power in the inverter, a power monitoring unit would be provided. This unit would monitor the grid power parameters and inverter power parameters and adjust them accordingly for proper synchronization. Inverter AC power quality would be continuously monitored and controlled to within the following limits:

- +10% to -18% voltage tolerance for normal operation once connected
- $\pm 3\%$ voltage tolerance during synchronization
- +1% to -5% frequency tolerance
- < 3% total harmonic distortion with no single harmonic > 2%
- Power factor range - 0.8 LG, 0.8 LD

C) Maximum Power Point Tracker (MPPT)

PV modules give maximum power output at a single set of generated voltage and generated current. MPPT is microprocessor-based unit and for a given radiation level it will track the maximum power point of the module to get maximum power output from the modules.

Terminal box is a part of a PV module, from which electrical output is taken. Each PV module would be provided with three by-pass diodes in the terminal box

B.2 Do no harm or Impact test of the project activity>>

There are social, environmental, economic and technological benefits which contribute to sustainable development.

Social benefits:

- The project activity will lead to the development of supporting infrastructure such as road network etc., in the wind park location, the access to which is also provided to the local population.
- The project activity will lead to alleviation of poverty by establishing direct and indirect benefits through employment generation and improved economic activities by strengthening of local grid of the state electricity utility.
- Use of a renewable source of energy reduces the dependence on imported fossil fuels and associated price variation thereby leading to increased energy security.

Environmental benefits:

- The project activity employs renewable energy source for electricity generation instead of fossil fuel- based electricity generation which would have emitted gaseous, liquid and/or solid effluents/wastes.
- Being a renewable resource, using wind energy to generate electricity contributes to resource conservation. Thus, the project causes no negative impact on the surrounding environment and contributes to environmental well-being.

Economic benefits:

- The project activity requires temporary and permanent, skilled and semi-skilled manpower at the wind park; this will create additional employment opportunities in the region.
- The generated electricity will be fed into the Indian grid through local grid, thereby improving the grid frequency and availability of electricity to the local consumers (villagers & sub- urban habitants) which will provide new opportunities for industries and economic activities to be setup in the area thereby resulting in greater local employment, ultimately leading to overall development.

United Nations Sustainable Development Goals:

The project activity generates electrical power using Solar energy, which is generated from windmills, thereby displacing non-renewable fossil resources resulting to sustainable, economic and environmental development. In the absence of the project activity equivalent amount of power generation would have taken place through fossil fuel dominated power generating stations. Thus, the renewable energy generation from project activity will result in reduction of the greenhouse gas emissions.

SDG Goals	Description
Goal 7 	This Solar energy project will generate clean electricity at a more accessible price for consumers. By utilizing a renewable resource, the project contributes to a growing share of clean energy sources in the global energy mix, ultimately reducing reliance on fossil fuels.
Goal 8 	This Solar project strengthens the local community by generating employment for the locals wherein a safe and inclusive work environment that empowers women is promoted.
Goal 13 	This 50 MW Solar power project meets the SDG 13 goal by displacing fossil fuel-based energy generation by producing clean energy. This project is expected to reduce 554,625 tCO ₂ emissions per year.

B.3. Baseline Emissions>>>

The baseline scenario identified at the PCN stage of the project activity is:

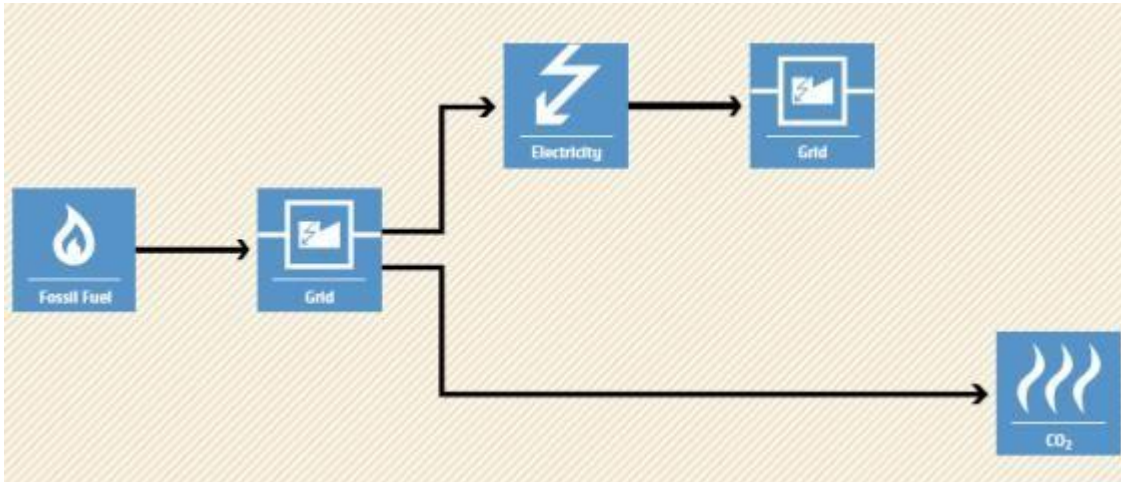
In the absence of the project activity, the equivalent amount of electricity would have been generated from fossil fuel-based power plants and exported to the regional grid (which is connected to the unified Indian Grid system) as national grid is predominantly sourcing from fossil fuel-based

© Universal CO2 Emission And Offset Registry Private Ltd

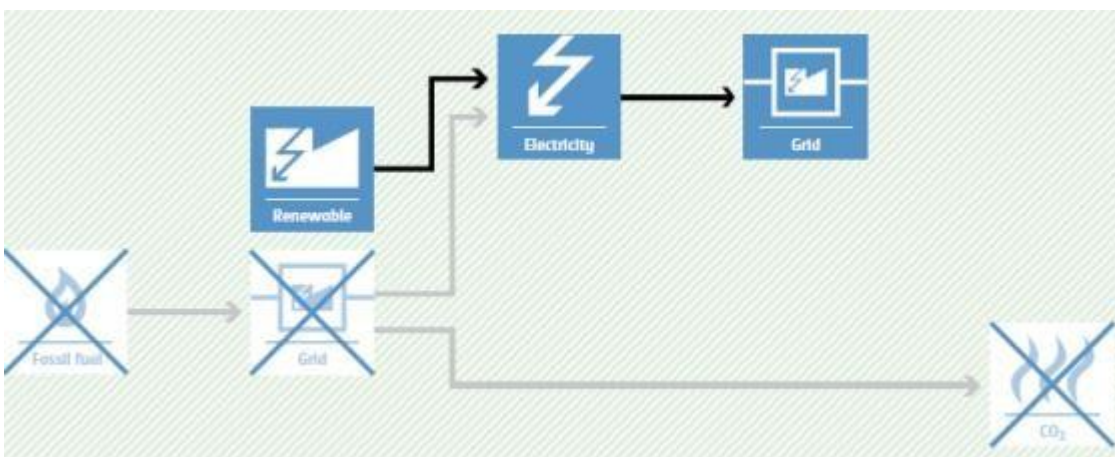
power plants. Hence, baseline scenario of the project activity is the grid-based electricity system, which is also the pre-project scenario.

Schematic diagram showing the baseline scenario:

Baseline Scenario:



Project Scenario:



B.4. Debundling>>

This project is not a debundled component of a larger registered carbon offset project activity.

SECTION C. Application of methodologies and standardized baselines

C.1. References to methodologies and standardized baselines >>

SECTORAL SCOPE – 01 Energy industries (Renewable/Non-renewable sources)

TYPE I- Renewable Energy Projects

CATEGORY- ACM0002., Consolidated baseline methodology for grid-connected electricity generation from renewable sources -Version 22.0

C.2. Applicability of methodologies and standardized baselines >>

The project activity involves generation of grid connected electricity from the construction and operation of a new Solar power project for selling it to grid. The project activity has an Operational capacity of 50 MW which will qualify for a large-scale project activity. The project status corresponds to the methodology ACM0002 version 22, and applicability of methodology is discussed below.

Applicability Criteria.	Applicability status
1) This methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Install a Greenfield power plant; (b) Involve a capacity addition to (an) existing plant(s); (c) Involve a retrofit of (an) existing operating plant(s)/unit(s); (d) Involve a rehabilitation of (an) existing plant(s)/unit(s), or (e) Involve a replacement of (an) existing plant(s)/unit(s). (f) Install a Greenfield power plant together with a grid-connected Greenfield pumped storage power plant. The greenfield power plant may be directly connected to the PSP or connected to the PSP through the grid.	The proposed project involves establishing a new grid-connected renewable Solar power plant, confirming to the specified criteria.
2) In case the project activity involves the integration of a BESS, the methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Integrate BESS with a Greenfield power plant; (b) Integrate a BESS together with implementing a capacity addition to (an) existing solar photovoltaic or wind power plant(s)/unit(s); (c) Integrate a BESS to (an) existing solar photovoltaic or wind power plant(s)/unit(s) without implementing any other changes to the existing plant(s); (d) Integrate a BESS together with implementing a retrofit of (an) existing solar photovoltaic or wind power	The project entails installing a new grid-connected renewable Solar power project without the integration of a Battery Energy Storage System (BESS). Therefore, this condition does not apply to the project activity.

plant(s)/unit(s). (e) Integrate a BESS together with a Greenfield power plant that is operating in coordination with a PSP. The BESS is located at site of the greenfield renewable power plant.	
3)The methodology is applicable under the following conditions: (a) Hydro power plant/unit with or without reservoir, wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit; (b) In the case of capacity additions, retrofits, rehabilitations or replacements (except for wind, solar, wave or tidal power capacity addition projects) the existing plant/unit started commercial operation prior to the start of a minimum historical reference period of five years, used for the calculation of baseline emissions and defined in the baseline emission section, and no capacity expansion, retrofit, or rehabilitation of the plant/unit has been undertaken between the start of this minimum historical reference period and the implementation of the project activity; (c) In case of Greenfield project activities applicable under paragraph 7(a) above, the project participants shall demonstrate that the BESS was an integral part of the design of the renewable energy project activity (e.g., by referring to feasibility studies or investment decision documents); (d) The BESS should be charged with electricity generated from the associated renewable energy power plant(s). Only during exigencies² may the BESS be charged with electricity from the grid or a fossil fuel electricity generator. In such cases, the corresponding GHG emissions shall be accounted for as project emissions following the requirements under section 5.4.4 below. The charging using the grid or using fossil fuel electricity generator should not amount to more than 2 per cent of the electricity generated by the project renewable energy plant during a monitoring period. During the time periods (e.g., week(s), months(s)) when the BESS consumes more than 2 per cent of the electricity for charging, the project participant shall not be entitled to issuance of the certified emission reductions for the concerned periods of the monitoring period. (e) In case the project activity involves PSP, the PSP shall utilize the electricity generated from the renewable energy power plant(s) that is operating in coordination with the PSP during pumping mode	The proposed project involves installing new Solar power plants without integrating a Battery Energy Storage System (BESS). Thus, the mentioned criterion does not apply
4)In case of hydro power plants, one of the following conditions shall apply: a)The project activity is implemented in an existing single or multiple reservoirs, with no change in the volume of any of reservoirs; or	The proposed project involves the installation of Solar power plants/units. Hence, the mentioned criterion is not applicable.

b)The project activity is implemented in an existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density calculated using equation (7) is greater than 4 W/m²; or c)The project activity results in new single or multiple reservoirs and the power density calculate equation (7), is greater than 4 W/m². d)The project activity is an integrated hydro power project involving multiple reservoirs, where the power density of any of the reservoirs, calculated using equation (7), is lower than or equal to 4 W/m², all of the following conditions shall apply. (i)The power density calculated using the total installed capacity of the integrated project, as per equation (8), is greater than 4 W/m²; (ii) Water flow between reservoirs is not used by any other hydropower unit which is not a part of the project activity; (iii)Installed capacity of the power plant(s) with power density lower than or equal to 4 W/m² are: a) Lower than or equal to 15 MW; and b) Less than 10 per cent of the total installed capacity of integrated hydro power project.	
5)In the case of integrated hydro power projects, project proponent shall: a)Demonstrate that water flow from upstream power plants/units spill directly to the downstream reservoir and that collectively constitute to the generation capacity of the integrated hydro power project; or b)Provide an analysis of the water balance covering the water fed to power units, with all possible combinations of reservoirs and without the construction of reservoirs. The purpose of water balance is to demonstrate the requirement of specific combination of reservoirs constructed under CDM project activity for the optimization of power output. This demonstration has to be carried out in the specific scenario of water availability indifferent seasons to optimize the water flow at the inlet of power units. Therefore, this water balance will take into account seasonal flows from river, tributaries (if any), and rainfall for minimum five years prior to implementation of CDM project activity.	The proposed project activity involves the installation of Solar power plants/units. Therefore, the mentioned criteria are not applicable.
6) In the case of PSP, the project participants shall demonstrate in the PDD that the project is not using water which would have been used to generate electricity in the baseline.	The proposed project activity involves installing Solar power plants/units. Therefore, the specified criteria are not applicable.
7)The methodology is not applicable to: a) Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity,	The proposed project activity involves installing Solar power plants/units. Therefore, the specified

since in this case the baseline may be the continued use of fossil fuels at the site; b) Biomass-fired power plants;	criteria are not applicable.
8)In the case of retrofits, rehabilitations, replacements, or capacity additions, this methodology is only applicable if the most plausible baseline scenario, as a result of the identification of baseline scenario, is “the continuation of the current situation, that is to use the power generation equipment that was already in use prior to the implementation of the project activity and undertaking business as usual maintenance	The proposed project activity involves installing solar power plants/units. Therefore, the specified criteria are not applicable.

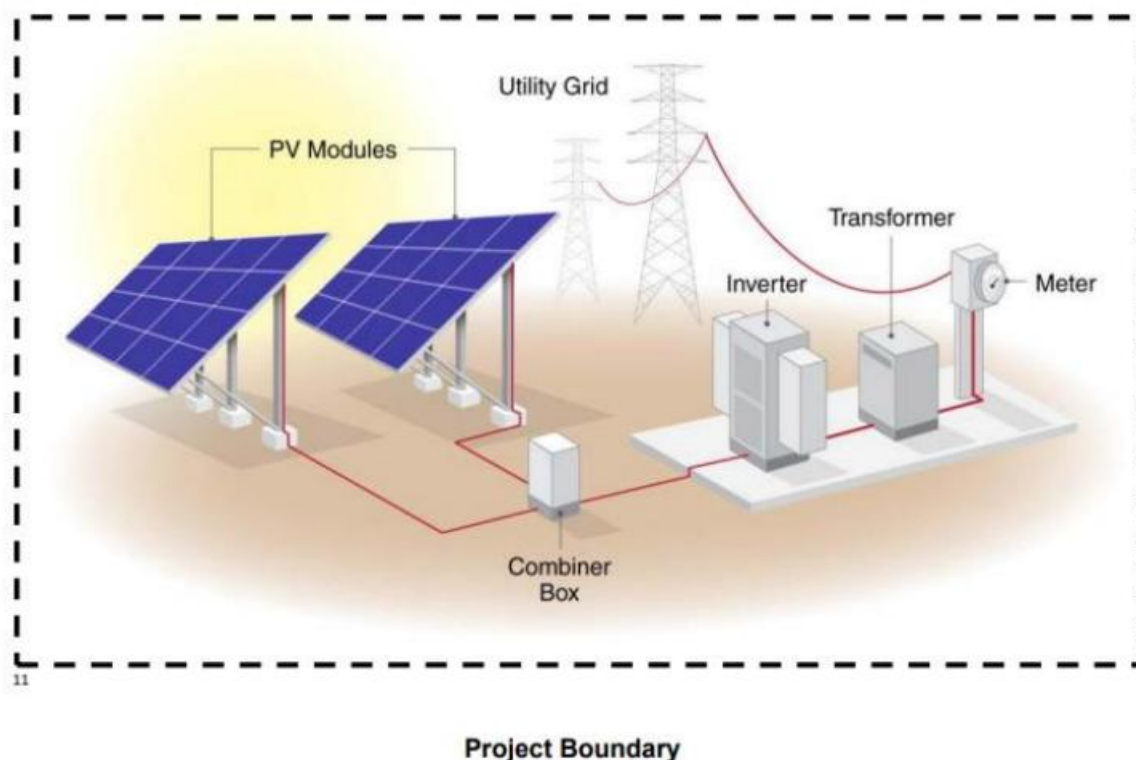
C.3 Applicability of double counting emission reductions >>

There is no double accounting of emission reductions in the project activity due to the following reasons:

- Project is uniquely identifiable based on its location coordinates,
- Project has dedicated commissioning certificate and connection point,
- Project is associated with energy meters which are dedicated to the consumption point for project developer.

C.4. Project boundary, sources and greenhouse gases (GHGs)>>

As per applicable methodology, the spatial extent of the project boundary includes the project power plant, and all power plants connected physically to the electricity system that the project power plant is connected to. Hence, the project boundary includes the project site where the power plant has been installed, associated power evacuation infrastructure, energy metering points, switch yards and other civil constructs and connected to the Indian grid of Telangana.



The baseline study of the Indian grid shows that the main sources of GHG emissions under the baseline scenario are CO₂ emissions from the conventional power generating systems. Other emissions are that of CH₄ and N₂O but both emissions have been excluded for simplification. The project activity generates

Source		GHGs	Included?	Justification/Explanation
Baseline scenario	Grid connected electricity generation	CO ₂	Yes	In the baseline scenario, the electricity would have been sourced from the Indian grid which in turn would be connected to fossil fuel fired power plants which emit CO ₂ .
		CH ₄	No	No methane is expected to be emitted.
		N ₂ O	No	No nitrous oxide is expected to be emitted.
Project Scenario	Greenfield wind energy	CO ₂	No	The project activity does not emit any emissions.
		CH ₄	No	No methane is expected to be emitted.

	conversion system	N ₂ O	No	No nitrous oxide is expected to be emitted.
--	-------------------	------------------	----	---

C.5. Establishment and description of baseline scenario (UCR Protocol) >>

As per the approved consolidated methodology ACM0002, version - 22, if the project activity is the installation of a new grid-connected renewable power plant/unit, the baseline scenario is the following:

“The baseline scenario is that the electricity delivered to the grid by the project activity would have otherwise, been generated by the operation of grid-connected power plants and by the addition of new generation sources into the grid”

The project activity involves setting up of a new grid connected Wind power plant to harness the green power from wind energy. In the absence of the project activity, the equivalent amount of power would have been supplied by the Indian grid, which is fed mainly by fossil fuel fired plants. The power produced at grid from the other conventional sources which are predominantly fossil fuel based. Hence, the baseline for the project activity is the equivalent amount of power produced at the Indian grid.

As per approved consolidated methodology ACM0002, version 22.0, emission reduction is estimated as difference between the baseline emission and project emission after factoring into leakage

Emission reductions are calculated as per methodology ACM0002, Version 22.0 Equation 17:

$$ER_y = BE_y - PE_y \quad (\text{Eq. 1})$$

Where,

ER_y = Emissions reductions in year y (t CO₂)

BE_y = Baseline emissions in year y (t CO₂)

PE_y = Project emissions in year y (t CO₂)

Baseline Emissions

The baseline emissions as per methodology ACM0002, Version 22.0, para 57; encompass solely the CO₂ emissions stemming from electricity generation in power plants displaced by the project activity. The methodology operates on the assumption that any electricity generation exceeding baseline levels would have originated from established grid-connected power plants and the integration of new grid-connected power plants.

The Baseline emissions as per methodology ACM0002, Version 22.0 Equation 17 in year y can be calculated as follows:

$$BE_y = EG_{PJ, y} * EF_{grid, CM, y}$$

Where:

BE_y = Baseline emissions in year y (tCO_2/yr)

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)

$EF_{grid,y}$ = Grid Emission factor in year y (tCO_2/MWh)

Since the project activity is the installation of a new grid connected renewable power plant (green field project), hence, $EG_{PJ,y}$ has been calculated as :

$$EG_{PJ,y} = EG_{facility,y}$$

Where:

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)

$EG_{facility,y}$ = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr)

A "grid emission factor" denotes the CO_2 emission factor (measured in tCO_2/MWh) associated with each unit of electricity supplied by an electricity system. The UCR suggests employing an emission factor of 0.9¹ from 2013 to 2023 and Emission Factor of 0.757 tCO_2/MWh for 2024.

Project Emission:

Regarding project emissions, ACM0002 version 22.0 specifies that only emissions related to fossil fuel combustion, emissions from the operation of geothermal power plants due to the release of non-condensable gases, and emissions from water reservoirs of hydroelectric plants should be taken into account. Since the project involves a wind power project, emissions from renewable energy plants are negligible

Hence ($PE_y = 0$).

Leakage Emission:

Leakage, as outlined in ACM0002 version 22.0, para 5.6, is considered to be zero as there is no transfer of energy-generating equipment in the project activity

Hence ($LE_y = 0$).

While the actual emission reduction achieved during the initial crediting period will be submitted during the first monitoring and verification, an ex-ante estimation is provided for reference.

Estimated Annual or Total baseline emission reductions (BE_y) = 554,625 CoUs /year (554,625 $tCO_{2eq}/year$)

¹As per [UCR CoU Standard Update: 2024 Vintage UCR Indian Grid Emission Factor Announced | by Universal Carbon Registry | Jan, 2025 | Medium](#)

Year	Quantity of net electricity generation supplied by the project activity to the grid in year y	Quantity of net electricity generation supplied by the project activity to the grid in year y	Emission Factor	Baseline Emissions	Project emissions or actual net GHG removals by sink	Leakage	Emission reductions or net anthropogenic GHG removals by sinks
	[KWh]	[MWh]	(tCO2e/MWh)	(tCO2e)	(tCO2e)	(tCO2e)	(tCO2e)
	[EGfacility, y]	[EGfacility, y]	[EFy]	[Bey]=[EGfacility, y]* [EFy]	[PEy]	[LEy]	[ERy]=[Bey]-[PEy]-[LEy]
Year 1	6204681.6	6204.6816	0.9	5584.21344	0	0	5584
Year 2	96551244	96551.244	0.9	86896.1196	0	0	86896
Year 3	97894848	97894.848	0.9	88105.3632	0	0	88105
Year 4	78047556	78047.556	0.9	70242.8004	0	0	70243
Year 5	73248828	73248.828	0.9	65923.9452	0	0	65924
Year 6	92955684	92955.684	0.9	83660.1156	0	0	83660
Year 7	91639396	91639.396	0.9	82475.4564	0	0	82475
Year 8	94765000	94765	0.757	71737.105	0	0	71737
Total		631307.2376					554625

C.6. Prior History>>

The project activity is a ground mounted Large -scale solar project and this project was never applied under any other GHG mechanism prior to this registration with UCR. Also, the capacity or the total project has not been applied for any other GHG mechanism. Hence project will not cause double accounting of carbon credits (i.e., COUs)

C.7. Monitoring period number and duration>>

First Issuance Period : 02/11/2017 to 31/12/2024 (07 years, 10 months, 18 days)
Monitoring Period : 02/11/2017 to 31/12/2024

C.8. Changes to start date of crediting period >>

There is no change in the start date of the crediting period.

C.9. Permanent changes from PCN monitoring plan, applied methodology or applied standardized baseline >>

There are no permanent changes from the registered PCN monitoring plan and applied methodology.

C.10. Monitoring plan>>

Data and Parameters available at validation (ex-ante values):

Data / Parameter:	EFGrid,y
Data unit:	tCO ₂ /MWh
Description:	A "grid emission factor" refers to a CO ₂ emission factor (tCO ₂ /MWh) which will be associated with each unit of electricity provided by an electricity system. The UCR recommends an emission factor of 0.9 tCO ₂ /MWh for the for the 2013 - 2023years and 0.757 tCO ₂ /MWh for year 2024.
Source of data:	UCR CoU Standard Update: 2024 Vintage UCR Indian Grid Emission Factor Announced by Universal Carbon Registry Jan, 2025 Medium
Measurement procedures (if any):	-
Monitoring frequency:	Ex-ante fixed parameter
QA/QC procedures:	For the calculation of Emission Factor of the grid
Any comment:	

Data and Parameters to be monitored

Data/Parameter	EGpj,y
Data unit	MWh
Description	“Form B” -Electricity Generated by Wind farm’ prepared & issued by SLDC
Measurement methods and procedures	- All the cluster meters and sub-station meters (main & check meters) are electronic and two-way (bi-directional) meters that measure both export and import of electricity and provide net electricity exported to the grid. -All the cluster meters and sub-station meters (main & check meters) measure the electricity (export & Import) on continuous basis and are recorded by state utility on monthly basis. Cross-sectional Data Type: Measured Monitoring equipment: Energy Meters are used for

	monitoring Archiving Policy: Electronic Calibration frequency: Once in 5 years ² (considered as per provision of CEA India). The net electricity generated by the project activity will be calculated Meter
Value Applied	631,307 MWh
Monitoring frequency	Monthly The net energy exported to the grid is measured every month using calibrated energy meter by the State Electricity Board authorities in the presence of the project implementer or its representatives. The meter/s shall be jointly inspected and sealed by authorised representatives of the company and the state utility. Measuring procedure: Will be measured by an export-import energy meter. The net electricity exported by the project plant would either be directly sourced as a measured parameter or be calculated by deducting the amount of imported electricity from the total amount of exported electricity.
QA/QC procedures	Continuous monitoring, hourly measurement monthly recording. Tri-vector (TVM)/ABT energy meters with accuracy class 0.2s.
Purpose of data	For baseline emission calculations

² https://cea.nic.in/wp-content/uploads/2020/02/meter_reg.pdf

Appendix> Calibration details

Company Name	Plant Capacity (MW)	Location	Commissioning Date	Calibration date	Calibration validity	Calibration delay
50 MW Solar Project by Sunshakti Solar Power Projects Pvt Ltd in Telangana	50 MW	Pedayyipally	02-11-2017	18 -05-2023 27-12-2024	18-05-2028 27-12-2029	01-11-2017 to 17-05-2023

Details of New Meters

During the calibration of ABT meters on 21.05.2024, it was observed that the backlight of the LCD was not working, and the LCD screen was fading in all Main, Check, and Standby energy meters. Due to this issue, it had become difficult to record JMR (Joint Meter Readings), and the fading LCD screen was also resulting in incorrect billing. Additionally, a few links in the TTB were found to have failed. For this reason, the meter was replaced on 27.12.2024.

Old Meter

Meter Details	Main Meter	Check Meter	Stand by Meter
Sl. No.	APX01518	APX01519	APX01520
Make	Secure	Secure	Secure
Accuracy	Class: 0.2 s	Class: 0.2 s	Class: 0.2 s

New Meter

Meter Type	Serial No.
Main Meter	APZ02558
Check Meter	APZ02559
Stand By Meter	APZ02560

There was a calibration delay from 01-11-2017 to 17-05-2023 in the monitoring period. To address this, an error factor has been applied to the net export values for the delay period since the meters were not calibrated as per the required frequency. According to VVS guidelines, an error factor of " $\pm 0.2\%$ " should be applied separately to both export and import values. A conservative approach has been adopted. To account for potential errors in both export and import, a cumulative error factor of " -0.4% " has been applied to the net electricity generation for the delay period.

Meter Photographs

